

A Point-Transitive Kirkman Triple System of Order 9: Disproof of Conjecture 00000008420 (Transitive-KTS Clause)

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Abstract

Conjecture 00000008420 asserts, among other clauses, that “the smallest order of a KTS with a transitive automorphism is the point-transitive type of order 15”. We exhibit an explicit counterexample: the line system of the affine plane $\text{AG}(2, 3)$ is a resolvable Steiner triple system of order 9 (a $\text{KTS}(9)$) whose translation group $\mathbb{F}_3^2$ acts transitively on the points. Since $9 < 15$, the clause is false. The construction is verified by independent enumeration (`reproduce.py`) and by an axiom-free Lean 4 formalization in which all statements are proved by `decide` over explicit finite enumerations.

1 Scope of the disproof

The conjecture under attack is a conjunction of four clauses. We refute only the third one:

“the smallest order of a KTS with a transitive automorphism is the point-transitive type of order 15”.

The other clauses are consistent with our counterexample and no claim is made about them: the order $9 \equiv 3 \pmod{6}$ (clause 1), and the number of parallel classes of the system below is exactly $4 = (9 - 1)/2$ (clause 2). Clause 4 (no KTS with cyclic automorphism group of prime order) is untouched.

2 Definitions

Definition 1 (Steiner triple system). *A Steiner triple system $\text{STS}(n)$ is a pair $(P, \mathcal{L})$ where P is a set of n points and $\mathcal{L}$ is a collection of 3-element subsets of P (blocks) such that every pair of distinct points lies in exactly one block.*

Definition 2 (Parallel class; resolvability). *Let $(P, \mathcal{L})$ be an STS. A parallel class is a subset $\mathcal{C} \subseteq \mathcal{L}$ of pairwise disjoint blocks whose union is P . The system is resolvable if $\mathcal{L}$ is partitioned into parallel classes. A Kirkman triple system $\text{KTS}(n)$ is a resolvable $\text{STS}(n)$.*

Definition 3 (Automorphism; point-transitivity). *An automorphism of $(P, \mathcal{L})$ is a bijection $\alpha : P \rightarrow P$ with $\alpha(\mathcal{L}) = \mathcal{L}$, i.e. $\{\alpha(B) : B \in \mathcal{L}\} = \mathcal{L}$. The system is point-transitive if its automorphism group acts transitively on P .*

3 The construction: KTS(9) as the lines of AG(2, 3)

Let $\mathbb{F}_3 = \{0, 1, 2\}$ and let the point set be $P = \mathbb{F}_3^2$, so $|P| = 9$. Define 12 lines:

- for each slope $m \in \mathbb{F}_3$ and intercept $b \in \mathbb{F}_3$,

$$L_{m,b} = \{ (x, mx + b) : x \in \mathbb{F}_3 \},$$

giving 9 non-vertical lines;

- for each $c \in \mathbb{F}_3$, the vertical line

$$V_c = \{ (c, y) : y \in \mathbb{F}_3 \},$$

giving 3 further lines.

Table 1 lists all 12 lines explicitly; the index k used there ($k = 3m + b$ for the slope lines, $k = 9 + c$ for the vertical lines) is the index used by the Lean formalization and the script.

k	type	points	parallel class
0	$m=0, b=0$	(0,0) (1,0) (2,0)	C_0
1	$m=0, b=1$	(0,1) (1,1) (2,1)	C_0
2	$m=0, b=2$	(0,2) (1,2) (2,2)	C_0
3	$m=1, b=0$	(0,0) (1,1) (2,2)	C_1
4	$m=1, b=1$	(0,1) (1,2) (2,0)	C_1
5	$m=1, b=2$	(0,2) (1,0) (2,1)	C_1
6	$m=2, b=0$	(0,0) (1,2) (2,1)	C_2
7	$m=2, b=1$	(0,1) (1,0) (2,2)	C_2
8	$m=2, b=2$	(0,2) (1,1) (2,0)	C_2
9	vertical, $c=0$	(0,0) (0,1) (0,2)	C_3
10	vertical, $c=1$	(1,0) (1,1) (1,2)	C_3
11	vertical, $c=2$	(2,0) (2,1) (2,2)	C_3

Table 1: The 12 lines of the system and the partition into parallel classes.

4 Verification of the properties

Each property below was checked twice, independently: (i) by a dependency-free Python enumeration (`reproduce.py`) that builds the 12 lines explicitly and enumerates all 36 point-pairs, the 4 parallel classes, the 9 translations and all pairs of points; and (ii) by an axiom-free Lean 4 formalization (`lean4/Main.lean`, toolchain `leanprover/lean4:v4.33.1`) whose theorems `sts_property`, `resolvable`, `translation_invariant`, `transitive` and `refute` are proved by `decide` over the explicit finite enumerations and depend on no axioms (no `sorryAx`, no `Classical.choice`, not even `propext`).

4.1 Steiner triple system

There are $\binom{9}{2} = 36$ unordered pairs of distinct points. The 12 lines contain $12 \cdot \binom{3}{2} = 36$ point-pairs in total, and the enumeration of Table 1 shows that no two lines share more than one point, i.e. all

36 line-pairs are distinct. Hence every pair of distinct points lies in exactly one line: $(P, \mathcal{L})$ is an STS(9). (Equivalently: for each of the 81 ordered pairs (p, q) with $p \neq q$, exactly one line contains both; each point lies on exactly 4 lines.)

4.2 Resolvability

The four parallel classes are

$$C_0 = \{0, 1, 2\}, \quad C_1 = \{3, 4, 5\}, \quad C_2 = \{6, 7, 8\}, \quad C_3 = \{9, 10, 11\},$$

using the line indices of Table 1. Within each class the three lines are pairwise disjoint and their union is all of $\mathbb{F}_3^2$: this is the elementary fact that over $\mathbb{F}_3$ the lines of fixed slope m (respectively the vertical lines) partition the plane. The four classes are disjoint and cover all 12 lines, so the system is resolvable: it is a KTS(9). The number of parallel classes is $4 = (9 - 1)/2$, matching clause 2 of the conjecture.

4.3 Translations are automorphisms

For $v \in \mathbb{F}_3^2$ let $t_v(p) = p + v$ (addition in $\mathbb{F}_3^2$). If p lies on $L_{m,b}$ then $t_v(p)$ lies on $L_{m,b+mv_1+v_2}$; if p lies on V_c then $t_v(p)$ lies on V_{c+v_1} . Thus the image of a line under t_v is again a line; since t_v is a bijection of P , the map induced on the line set is a bijection of $\mathcal{L}$, i.e. t_v is an automorphism. This was verified by explicit enumeration for all 9 translations and all 12 lines (checking for each (v, k) that a line k' with $p \in L_k \Leftrightarrow t_v(p) \in L_{k'}$ for all p exists).

4.4 Point-transitivity

For all $p, q \in \mathbb{F}_3^2$ the translation t_{q-p} maps p to q :

$$t_{q-p}(p) = p + (q - p) = q.$$

Hence the automorphism group contains the translation subgroup $\{t_v : v \in \mathbb{F}_3^2\} \cong \mathbb{F}_3^2$ of order 9, which acts transitively on the 9 points. The system is point-transitive.

5 Conclusion

Theorem 1. *There exists a point-transitive Kirkman triple system of order 9, namely the line system of AG(2, 3).*

Theorem 2 (Disproof of clause 3 of conjecture 00000008420). *The smallest order of a KTS with a transitive automorphism is at most 9, and $9 < 15$. Therefore the clause “the smallest order of a KTS with a transitive automorphism is the point-transitive type of order 15” is false.*

Remark 1. *The full automorphism group of KTS(9) as constructed is $\text{AGL}(2, 3)$, of order $9 \cdot 48 = 432$; only the transitivity of its translation subgroup is used above.*